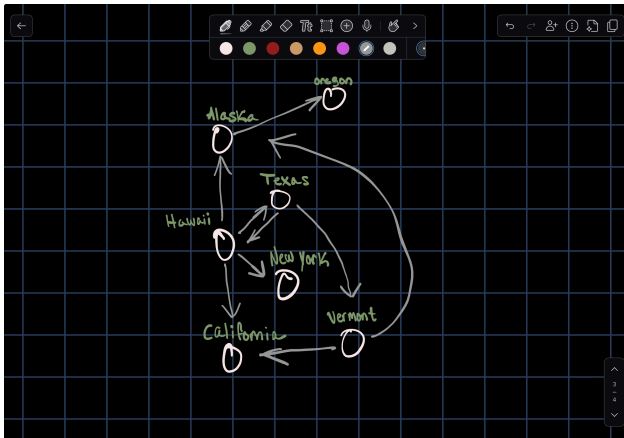


# CMSC204

## Kartchner

$V(\text{StateGraph}) = \{\text{Oregon, Alaska, Texas, Hawaii, Vermont, New York, California}\}$   
 $E(\text{StateGraph}) = \{(\text{Alaska, Oregon}), (\text{Hawaii, Alaska}), (\text{Hawaii, Texas}), (\text{Texas, Hawaii}), (\text{Hawaii, California}), (\text{Hawaii, New York}), (\text{Texas, Vermont}), (\text{Vermont, California}), (\text{Vermont, Alaska})\}$

1. Draw the StateGraph



1. Describe the graph pictured above, using the formal graph notation.

$V(\text{StateGraph}) = \{\text{Oregon, Alaska, Texas, Hawaii, Vermont, New York, California}\}$

$E(\text{StateGraph}) = \{(\text{Alaska, Oregon}), (\text{Hawaii, Alaska}), (\text{Hawaii, Texas}), (\text{Texas, Hawaii}), (\text{Hawaii, California}), (\text{Hawaii, New York}), (\text{Texas, Vermont}), (\text{Vermont, California}), (\text{Vermont, Alaska})\}$

2. a. Is there a path from Oregon to any other state in the graph? no. Its out degree is 0.

b. Is there a path from Hawaii to every other state in the graph? yes

c. From which state(s) in the graph is there a path to Hawaii?

texas.

3. a. Show the adjacency matrix that would describe the edges in the graph.  
Store the vertices in alphabetical order

States

AK  
 CA  
 HI  
 NY  
 OR  
 TX  
 VT

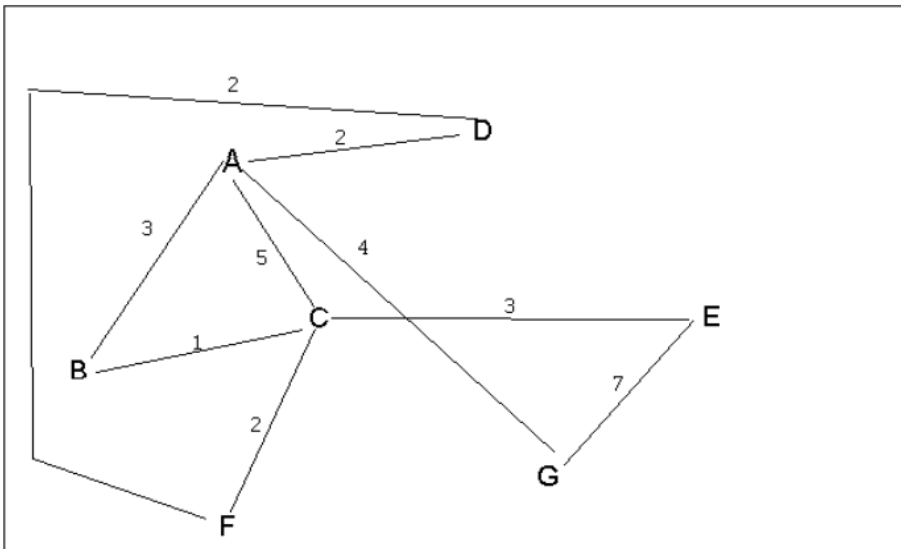
ADJacency MATRIX

|    | AK | CA | HI | NY | OR | TX | VT |
|----|----|----|----|----|----|----|----|
| AK | 0  | 0  | 0  | 0  | 1  | 0  | 0  |
| CA | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| HI | 1  | 1  | 0  | 1  | 0  | 1  | 0  |
| NY | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| OR | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| TX | 0  | 0  | 1  | 0  | 0  | 0  | 1  |
| VT | 1  | 1  | 0  | 0  | 0  | 0  | 0  |

Edge[1][7] = 1

3. b. Show the adjacency lists  
that would describe the edges in the graph

AK → OREGON  
 CA → null  
 HI → [AK, CA, NY, TX]  
 NY → null  
 OR → null  
 TX → [HI, VT]  
 VT → [AK, CA]

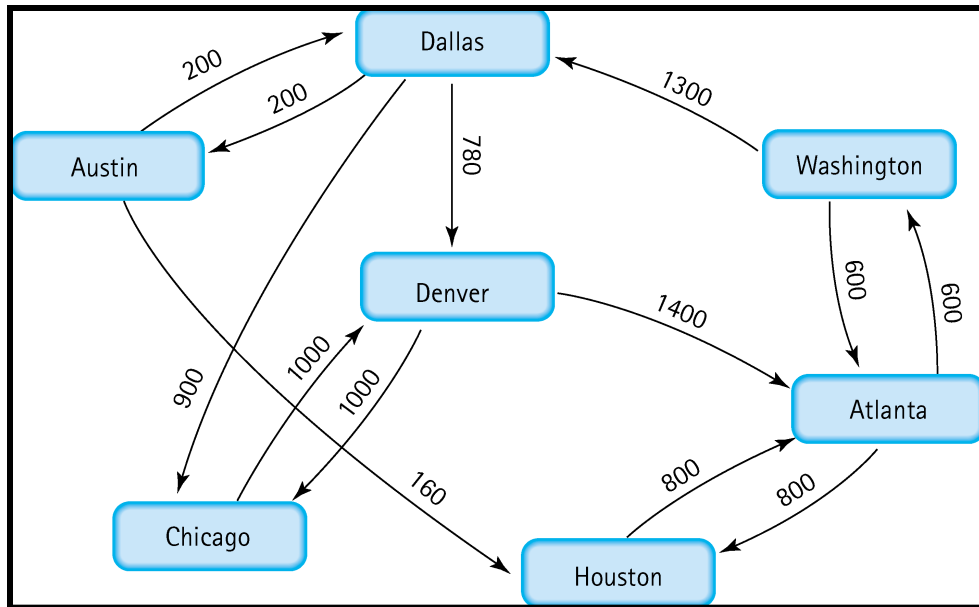


4 a. Which of the following lists the graph nodes in depth first order beginning with E?

- A) E, G, F, C, D, B, A
- B) G, A, E, C, B, F, D
- C) E, G, A, D, F, C, B**
- D) E, C, F, B, A, D, G

4 b. Which of the following lists the graph nodes in breadth first order beginning at F?

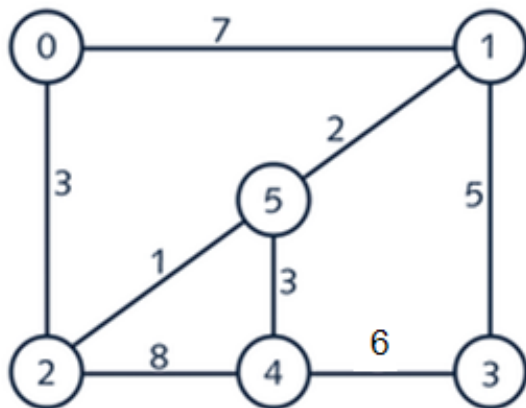
- A) F, C, D, A, B, E, G
- B) F, D, C, A, B, C, G
- C) F, C, D, B, G, A, E
- D) a, b, and c are all breadth first traversals**



5. Find the shortest distance from Atlanta to every other city

| City       | Distance                  |
|------------|---------------------------|
| Atlanta    | 0                         |
| Washington | 600                       |
| Houston    | 800                       |
| Dallas     | $\infty \rightarrow 1900$ |
| Denver     | $\infty \rightarrow 2680$ |
| Chicago    | $\infty \rightarrow 2800$ |
| Austin     | $\infty \rightarrow 2100$ |

6. Find the minimal spanning tree using Prim's algorithm. Use 0 as the source vertex . Show the steps.



MST

0 - 3 - 2 - 1 - 5

1 - 2 - 5

4 - 3 - 5

1 - 5 - 3

$\{0, 2, 5, 1, 4, 3\}$

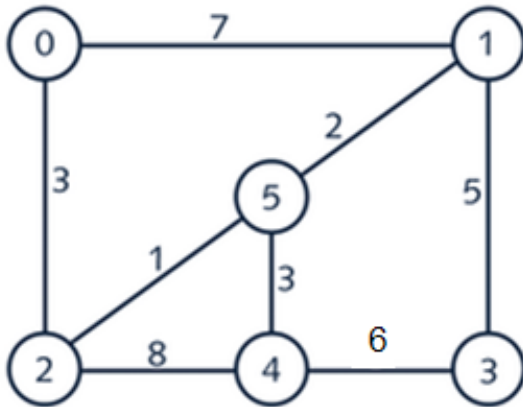
$3 + 1 + 2 + 3 + 5 = 14$

Total weight = 14

```

graph TD
    0 --- 2
    2 --- 5
    5 --- 1
    5 --- 4
    4 --- 3
  
```

7. Find the minimal spanning tree using Kruskal's algorithm. Show the weights in order and the steps.



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Kruskal

1 <sup>2</sup> 5 <sup>1</sup> 2 <sup>3</sup> 0

5 | 3

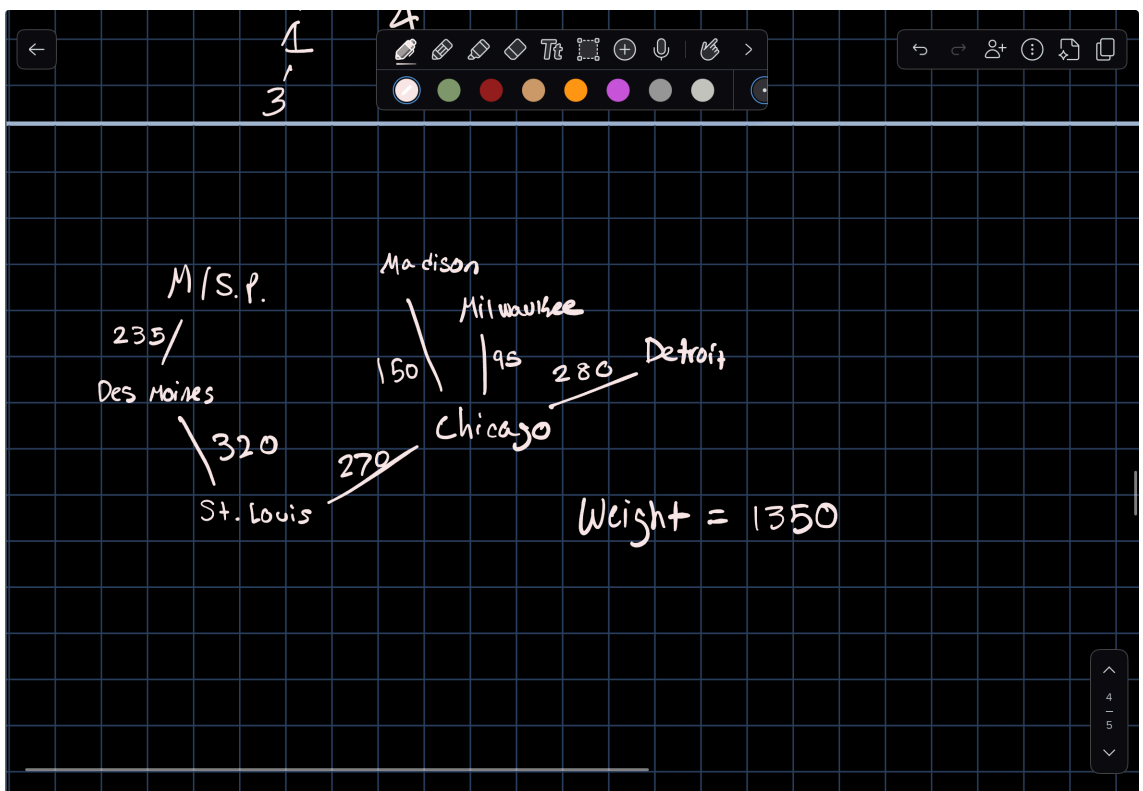
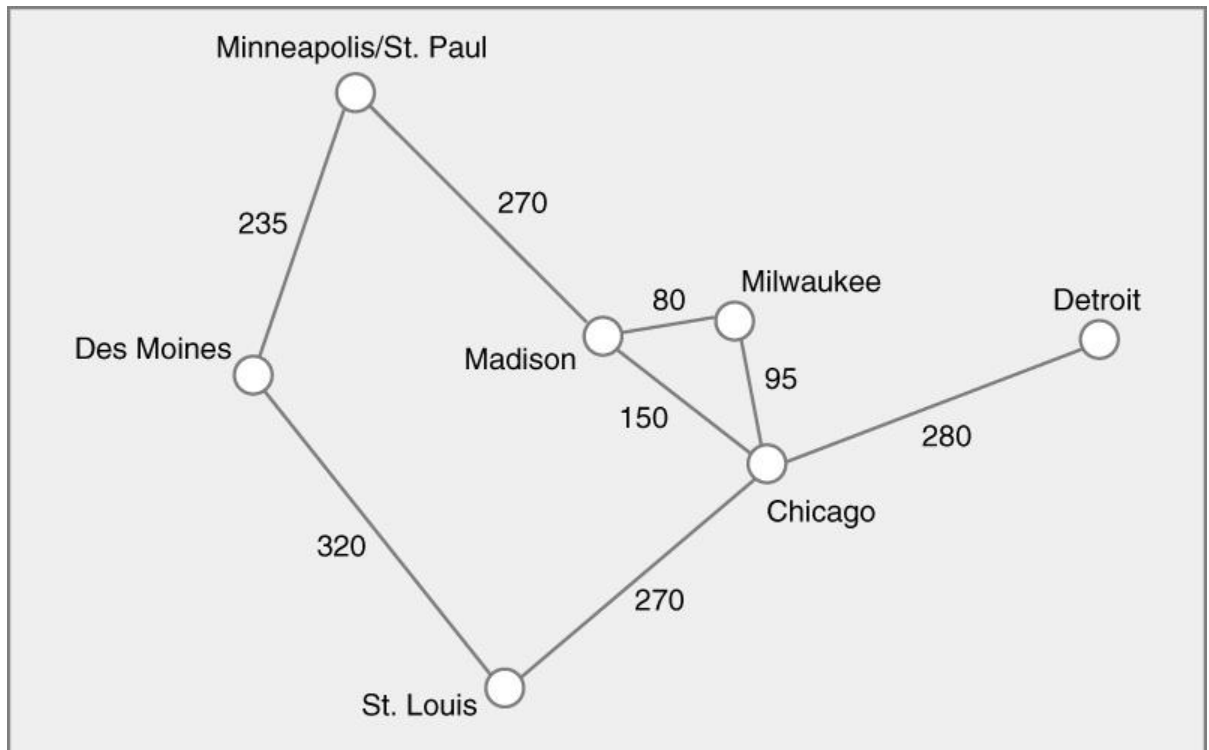
3 / 4

= 14

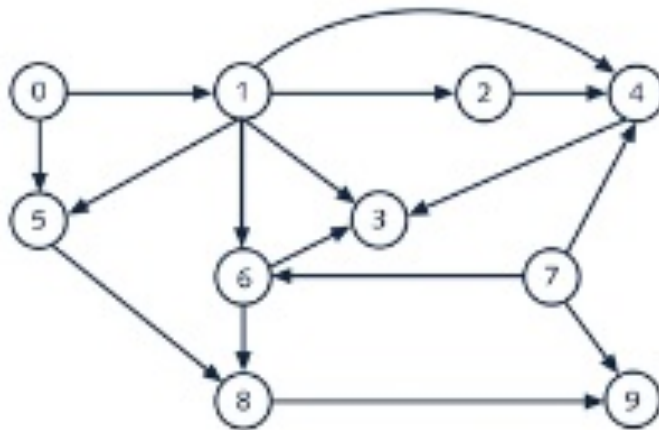
$N - 1$

$6 - 1 = 5$

8. Find the minimal spanning tree using the algorithm you prefer. Use Minneapolis/St. Paul as the source vertex



9. List the nodes of the graph in a breadth first topological ordering. Show the steps using arrays predCount, topologicalOrder and a queue



6 - 1 = 5

predCount

0 = 0

1 = ~~1~~ 0

2 = 1

3 = 3

4 = ~~3~~ 2

5 = 2 → 1

6 = ~~2~~ 1

7 = 0

8 = 2

9 = ~~2~~ 1

topologicalOrder

[0, 7, 1, 2, 5, 6, 3, 4, 8, 9]

Queue

[2, 5, 6]



10. List the nodes of the graph in a breadth first topological ordering.

